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Classification of motor imagery EEG signals using deep learning

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***Abstract*—Brain-computer interface (BCI) is a direct communication between the brain and the computer. The BCI system using an electroencephalogram (EEG), a non-invasive system, is used primarily for its high portability and practicality. This work aims to achieve the best approach to analyzing EEG signals by comparing both methods: the short-time Fourier transform (STFT) and the continuous wavelet transform (CWT). Both of these methods allow effective characterization of the signal in both time and frequency domains. Following the transformation through TFR, MI-based EEG signals are displayed as images. These images are then fed into a convolutional neural network (CNN) for feature extraction and classification. The assessment was performed using the BCI Competition 2003 dataset which includes two motor imagery classes from only one subject. Our findings demonstrated the CWT approach superior performance relative to the proposed STFT approach. In addition, the results show that the method using continuous wavelet transform with convolutional neural network can be comparable or better than the other state-of-the-art approaches.**

Keywords*—Brain-computer interface (BCI), Electroencephalogram (EEG), Short time Fourier transform (STFT), Continuous wavelet transform (CWT), Time-frequency representation (TFR), Motor Imagery (MI), Convolutional Neural Network (CNN).

I. Introduction

The Brain-Computer Interface (BCI) system has emerged as a human-computer communication method that translates neural activity as a control signal [1, 2]. It can directly translate human brain electrical activity i.e., Electroencephalogram (EEG) signals into external commands, without the involvement of peripheral nerves and muscles [3].

Most BCI studies are based on EEG signals since this procedure is non-invasive, inexpensive and easy to apply to collect information from the brain with sufficiently high temporal resolution. The EEG signal is recorded using multiple electrodes placed on the specific scalp areas.

Various EEG signals have been used in BCI systems, such as P300 potential [4], steady-state visual evoked potentials (SSVEP), and motor imagery (MI). Among these EEG signals, the MI signal is one of the most common signals, as it can be generated spontaneously without any stimulation. Motor imagery refers to the imagination of limb movements through the brain in a stationary state. The event-related synchronization (ERS) and event-related desynchronization (ERD) phenomena are accompanied by motor imagery [5]. ERS is manifested by the increase of mu and beta rhythm energy of ipsilateral sensory-motor cortex and the ERD is manifested by the decline of contralateral mu and beta rhythm energy [6].

Deep learning is the current research hotspot and has made outstanding progress in many research fields. Currently, Feng Li et al. [7] combined continuous wavelet transform (CWT) and simplified convolutional neural network (SCNN) to identify EEG signals for left and right-hand motor imagery and improve the performance of BCI systems. Y. Shen et al. [8] designs a new neural network, which can directly classify the unprocessed EEG data by using convolutional neural network (CNN) and long short-term memory (LSTM). Tabar and Halici used short-time Fourier transform (STFT) to extract features from EEG signals classified with a CNN-1D followed by a Stacked auto-encoder (SAE) [9].

In this paper, we propose a new input form, which uses two approaches wavelet transform and short time Fourier transform to convert multichannel EEG signals into two-dimensional time-frequency images, to obtain its comprehensive information, including both the time-frequency features and the relative position of the electrodes. Additionally, convolutional neural network is utilized as the classifier.

The remaining paper is structured as follows: Section II presents the dataset used, briefly explains the methods and relevant theories such as STFT, CWT, and CNN. The experimental results and discussion are depicted in Section III. Ultimately, section IV concludes this paper.

II. Methodology

A. Dataset description

The dataset we use is the BCI Competition 2003.This dataset was obtained from a normal individual (female, 25 years) [10]. Three of the bipolar EEG channels (C3, Cz, and C4) were used to record brain activity [11]. As illustrated in Fig. 1, the electrode locations correspond to the International 10-20 system.

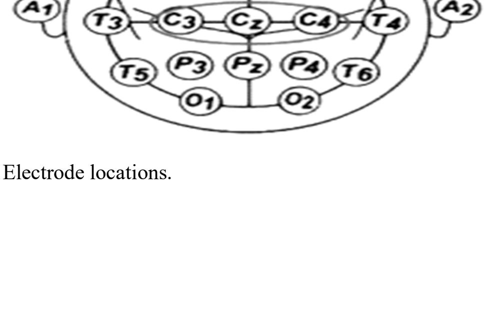

Fig. 1. Electrode locations.

There are 280 trials total in the experiment, 140 of which are for testing and 140 for training. Each trial lasts 9 seconds. The first two seconds were quiet, the trial started with an auditory stimulus at t=2s, and the first second was marked by a cross ('+'). Subsequently, at t=3s, a left or right arrow was shown as a cue, and the female was instructed to visualize moving her hands in that direction. The adaptive autoregressive parameters (AAR) of channels 1 (C3) and 3 (C4) served as the basis for the feedback. The AAR were merged with a discriminant analysis to create a single output parameter.

The dataset was bandpass filtered between 0.5 Hz and 30 Hz after being recorded at a sampling frequency of 128 Hz. There are 1152 sampling points for every trial. A single trial's paradigm is shown in Fig. 2.

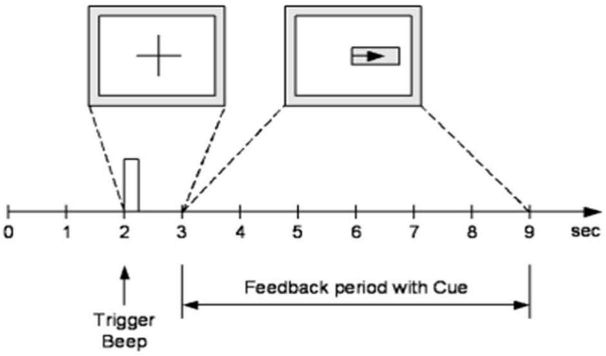

Fig. 2. Scheme of database paradigm.

In this study, we used only the C3 and C4 electrodes, where a previous study has shown that they contain most of the useful information for this particular application [12].

B. Proposed work

In this article, we employed two time frequency transformation (TFR) methods to compare their utilities towards completing the task of discrimination of motor imageries. These two TFR methods are STFT and CWT, which are widely popular for their remarkable properties. These time frequency (TF) analysis methods allow to accurately evaluate the non-stationary EEG signal in both time and frequency domains. They basically transform the one-dimensional EEG signals into two-dimensional time-frequency images. In this study, their utility is explored for the distinction of right hand (RH) and left hand (LH) MI tasks in a BCI system.

At first, the EEG signals of C3 and C4 channels are selected, the EEG data was bandpass filtered 8-30 Hz, and the raw data are filtered by a Butterworth FIR bandpass filter (order 1) to obtain the motor imagery EEG signals of 8–14 Hz and 15–30 Hz. Moreover, TF representation of the EEG signal is estimated using spectrograms and scalograms obtained through STFT and CWT, respectively. The scalogram obtained using CWT method is represented in Fig. 3. Similarly, Fig. 4 depicts the spectrogram obtained using STFT. Both STFT and CWT images were resized to 224×224×3. These were used as input to CNN model for feature extraction and classification purpose. The wavelet used for CWT is the analytic Morse wavelet as it has better time-frequency localization. For Morse wavelet symmetry parameter (gamma) and time bandwidth product were kept 3 and 60 respectively. The proposed model architecture is illustrated in Fig. 5.

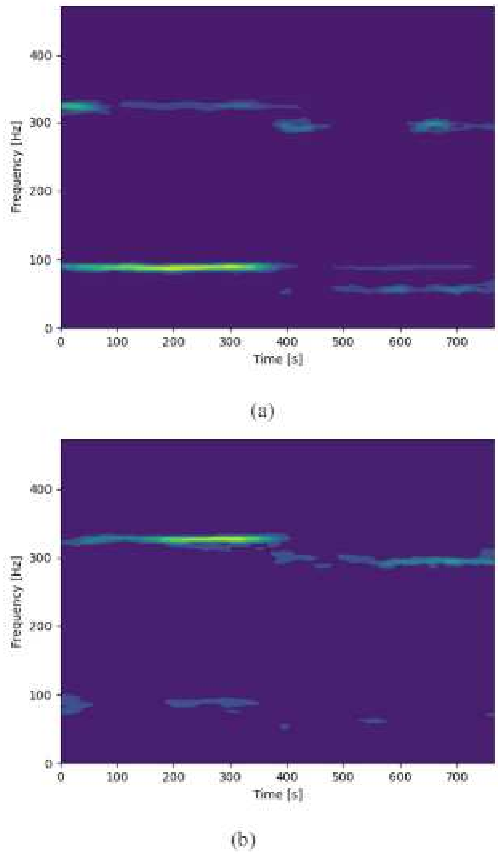

Fig. 3. Scalogram obtained using CWT for (a) LH and (b) RH respectively.

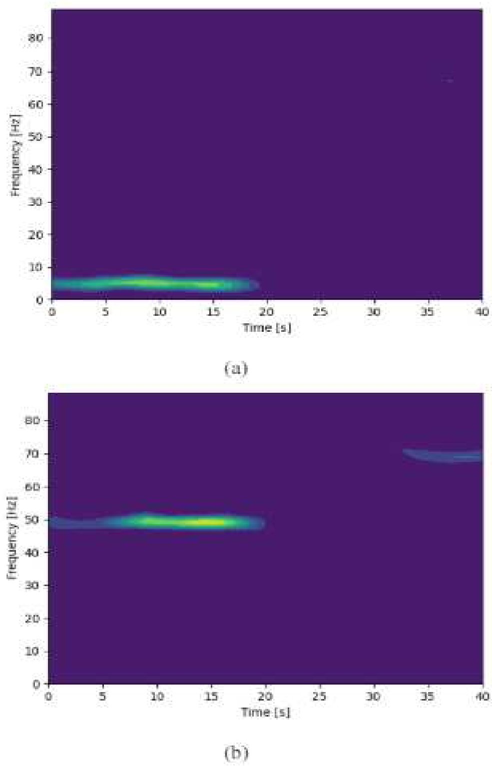

Fig. 4. Spectrogram obtained using STFT for (a) LH and (b) RH respectively.

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For training and evaluation, the two proposed methods were implemented in a Python environment using the "keras" library which is based on the Open Source "Jupyter" project.

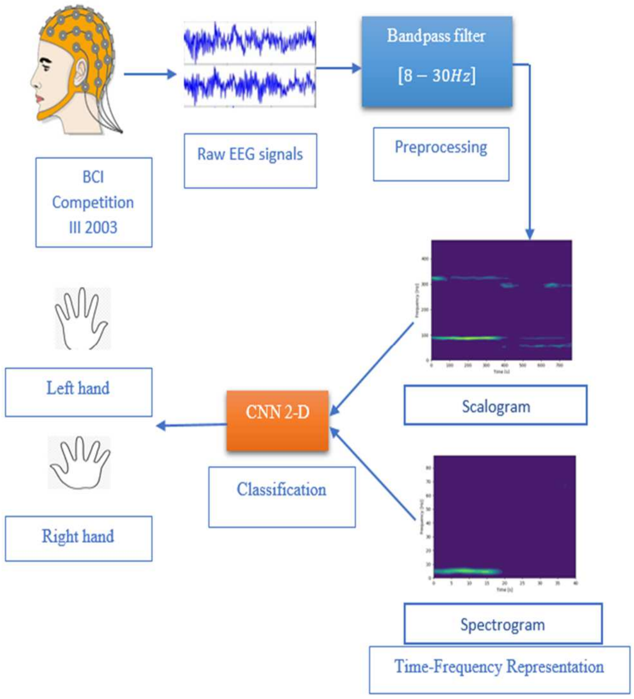

Fig. 5. Proposed methodology.

C. Short Time Fourier Transform (STFT)

Short-time Fourier transform (STFT) is a time-frequency analysis used for non-stationary signals analysis [13, 14]. It applies the Fourier transform to an extended temporal signal after dividing it into segments the same size as the window. STFT of signal $s(t)$ is $F(\tau,\omega)$.

$$F(\tau,\omega)=\int_{-\infty}^{+\infty} s(t)h(t-\tau)e^{-j\omega t}dt \quad (1)$$

where $h(t)$ is the windowing function or a low pass filter. The resultant spectrogram is magnitude squared of the STFT, given as Spectrogram $=|F(\tau,\omega)|^2$. The assumption is that $F(t)$ nearly stationary in the duration of the temporal window $h(t)$, and the length of the windows must be equal.

D. Continuous Wavelet Transform (CWT)

The advantage of non-stationarity and non-linearity in EEG signals is that they preserve signal spectrum information in addition to temporal frequency domain information. The CWT has a finite wave length and a mean value of zero. In EEG analysis, the scaling parameters and the time shifting are crucial factors. The definition of CWT is the signal's total time multiplied by time-shifted, scaled wavelet versions [15, 16]. The continuous wavelet transform (CWT) formula is as follows:

$$w_{(a,b)}[y(t)]=\frac{1}{\sqrt{a}}\int_{-\infty}^{\infty} y(t)\varphi^*\left(\frac{t\text{-}b}{a}\right)dt \quad (2)$$

Where $\varphi\left(\frac{t-b}{a}\right)$ is scaled and shifted version of the basis wavelet function $\varphi(t)$, a>0 is the scaling parameter that regulates the spread of the function, and b is the time shift parameter or the instant of time at which the signal needs to be analyzed. Thus, it is effectually identical to bandpass filtering of $y(t)$ at distinct scales.

E. Convolutional Neural Network (CNN)

Convolutional neural network (CNN) is a special form of deep neural networks. CNN consists of one input layer, one output layer and in between them are multiple convolutional, and dense layers. ConvNet is able to adopt the spatial and temporal information in an image. The objectives of convolution operation is to extract the features. CNN reduces the preprocessing part compared to traditional classification algorithms and hidden layers of CNN carry out the input data in a filtered form. Deep convolutional layers learn more complex features [17].

We propose a methodology for EEG classification to analyze the EEG images. This network has scalable high-performance architecture. Input image size is (224 × 224 × 3). The input images are convolved with filters in the convolution layers. In this paper, We use four convolutional layers, four ReLU, four pooling layers, one flattened layer and two fully connected layers. The first convolutional layer (conv-1) filters the input image with 16 kernels of size 4 x 4. The second convolutional layer (conv-2) takes the output of the first convolutional layer and filters with 10 kernels of size 3 x 3. The third convolutional layer (conv-3) takes the output of the second convolutional layer and filters with 10 kernels of size 3 x 3. The last convolutional layer (conv-4) takes the output of the third convolutional layer and filters with 64 kernels of size 4 x4. In each convolution layer, we find a maximum pooling layer of size 2 × 2. Max pooling layer minimizes the high dimensional input into smaller ones. Finally, a first fully connected layer (FC-5) is attached on the top of the convolutional layer with 45 neurons and the second fully connected layer (FC-6) was used the softmax activation function to compute the predicted labels. Our design of CNN architecture is shown in Fig. 6.

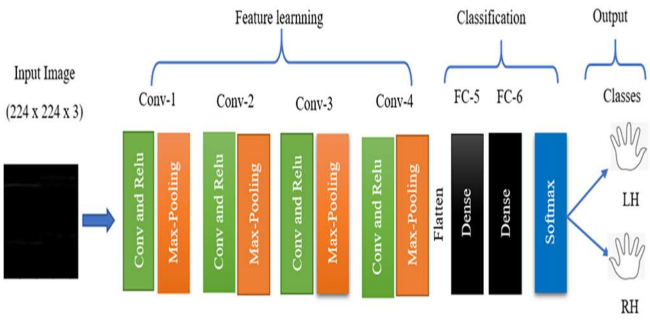

Fig. 6. CNN architecture.

We have perfected this model with 200 epochs. The batch size was 16. The model used both Adam optimizers and the sparse-categorical cross-entropy loss function with a learning rate of 0.0001.

III. Resultats and discussions

The performance criteria we used to evaluate the performance of the two models are the accuracy value and the kappa value. They are defined by the following equations.

$$Accuracy(\%)=\frac{TP+TN}{TP+TN+FP+FN}\times 100 \quad (3)$$

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Where:

- **TP (True Positive)** is the number of motor imagery tasks that are correctly detected.
- **TN (True Negative)** is the number of non-motor imagery tasks that are correctly classified as non-motor imagery tasks.
- **FP (False Positive)** is the number of motor imagery tasks that are incorrectly determined by a classifier.
- **FN (False Negative)** is the actual number of motor imagery tasks that are incorrectly assigned to other classes.

$$Kappa = \frac{acc - \frac{1}{N}}{1 - \frac{1}{N}} \quad (4)$$

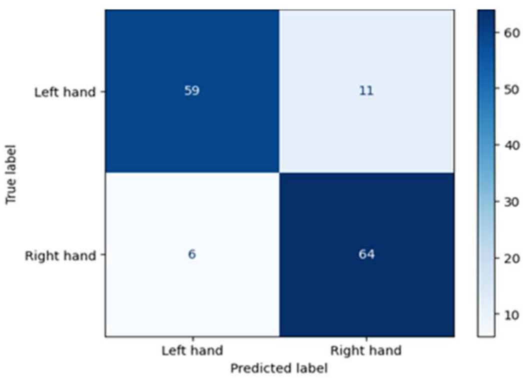
Where acc represents accuracy, N represents the number of categories, and N in this experiment is 2.

According to Table I, It can be noted the proposed CWT model and the proposed STFT reach their maximum values in terms of accuracy, respectively 87,86% and 85,71%. The kappa value achieved using STFT is 0.71 which is 0.04 lower than CWT. The performance of CWT is better in comparison to STFT for classification.

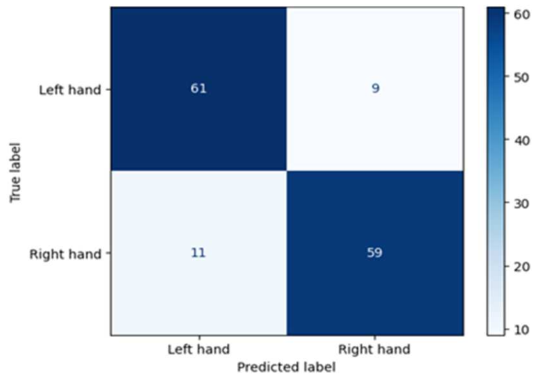
TABLE I. AVERAGE CLASSIFICATION RESULTS.

Parameters	CNN-2D	
	CWT	*STFT*
Max Accuracy (%)	87.86	85.71
Kappa value	0.75	0.71

Fig. 7 demonstrates the confusion matrices obtained using TFR methods. The diagonal elements demonstrate the number of points for which the predicted label is equal to the true label, while off-diagonal elements are those that are mislabeled by the classifier. The higher the diagonal values of the confusion matrix, the better, showing many correct predictions.



(a) CWT approach.



(b) STFT approach.

Fig. 7. Confusion matrices of the two approaches.

Table II, compares the classification accuracy of the proposed method with other existing methods. It is observed that using continuous wavelet transform with CNN, the proposed method provides better classification accuracy in detecting left and right hand motor imagery movements than others methods.

TABLE II. SUMMARY OF CLASSIFICATION ACCURACIES FOR THE INDICATED METHODS.

Authors	Year	Methods	Accuracy (%)
Darvishi et al. [18]	2007	CWT and ANFIS classifier	82.10
Yi et al. [19]	2012	Fisher classifier	86.30
Hu et al. [20]	2013	JR model and spectral powers	80.00
Jang et al. [21]	2016	STFT with K-Nearest Neighbor	83.57
Eslahi et al. [22]	2019	GA based optimization with FKNN, LDA	84.00
Kant et al. [23]	2019	CWT with Shannon entropy using SVM and KNN	86.40
Our approach		CWT with CNN	87.86

IV. CONCLUSION

In this work, we have presented a usage of CNN for MI EEG patterns, which incorporates 2D CNN structure with the CWT and STFT. Using two TF approaches for conversion of input EEG signals into images. The results obtained with the CWT technique are better than that obtained using the STFT method. This indicates that the CWT method provides a better representation of EEG signals in the TF domain as compared to the STFT method. In other words, the CWT images may have more distinctive features for EEG signals than the STFT method. The obtained results show that the proposed methodology is more effective for MI-tasks classification as compared to existing methods. As future works, we are working on developing and analyzing more effective neural networks schemes on MI BCI researches.

REFERENCES

[1] H. K. Lee and Y.-S. Choi, “A convolution neural networks scheme for classification of motor imagery EEG based on wavelet time-frequecy image,” *2018 Int. Conf. Inf. Netw. ICOIN*, pp. 906–909, Jan. 2018, doi: 10.1109/ICOIN.2018.8343254.

[2] D. J. McFarland and J. R. Wolpaw, “Brain-Computer Interfaces for Communication and Control,” *Commun. ACM*, vol. 54, no. 5, pp. 60–66, 2011, doi: 10.1145/1941487.1941506.

[3] N. F. Ince, F. Goksu, A. H. Tewfik, and S. Arica, "Adapting subject specific motor imagery EEG patterns in space–time–frequency for a brain computer interface," *Biomed. Signal Process. Control*, vol. 4, no. 3, pp. 236–246, Jul. 2009, doi: 10.1016/j.bspc.2009.03.005.

[4] L. da Silva-Sauer, L. Valero-Aguayo, A. de la Torre-Luque, R. Ron-Angevin, and S. Varona-Moya, "Concentration on performance with P300-based BCI systems: A matter of interface features," *Appl. Ergon.*, vol. 52, pp. 325–332, Jan. 2016, doi: 10.1016/j.apergo.2015.08.002.

[5] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," *Clin. Neurophysiol.*, vol. 110, no. 11, pp. 1842–1857, Nov. 1999, doi: 10.1016/S1388-2457(99)00141-8.

[6] Z. Tang, S. Sun, S. Zhang, Y. Chen, C. Li, and S. Chen, "A Brain-Machine Interface Based on ERD/ERS for an Upper-Limb Exoskeleton Control," *Sensors*, vol. 16, no. 12, Art. no. 12, Dec. 2016, doi: 10.3390/s16122050.

[7] F. Li, F. He, F. Wang, D. Zhang, Y. Xia, and X. Li, "A Novel Simplified Convolutional Neural Network Classification Algorithm of Motor Imagery EEG Signals Based on Deep Learning," *Appl. Sci.*, vol. 10, no. 5, Art. no. 5, Jan. 2020, doi: 10.3390/app10051605.

[8] Y. Shen, H. Lu, and J. Jia, "Classification of Motor Imagery EEG Signals with Deep Learning Models," in *Intelligence Science and Big Data Engineering*, Y. Sun, H. Lu, L. Zhang, J. Yang, and H. Huang, Eds., in Lecture Notes in Computer Science. Cham: Springer International Publishing, 2017, pp. 181–190. doi: 10.1007/978-3-319-67777-4_16.

[9] Y. R. Tabar and U. Halici, "A novel deep learning approach for classification of EEG motor imagery signals," *J. Neural Eng.*, vol. 14, no. 1, p. 016003, Nov. 2016, doi: 10.1088/1741-2560/14/1/016003.

[10] "BCI Competition II." https://www.bbci.de/competition/ii/ (accessed Sep. 10, 2023).

[11] S. K. Bashar and M. I. H. Bhuiyan, "Classification of motor imagery movements using multivariate empirical mode decomposition and short time Fourier transform based hybrid method," *Eng. Sci. Technol. Int. J.*, vol. 19, no. 3, pp. 1457–1464, Sep. 2016, doi: 10.1016/j.jestch.2016.04.009.

[12] P. Kant, S. H. Laskar, J. Hazarika, and R. Mahamune, "CWT Based Transfer Learning for Motor Imagery Classification for Brain computer Interfaces," *J. Neurosci. Methods*, vol. 345, p. 108886, Nov. 2020, doi: 10.1016/j.jneumeth.2020.108886.

[13] J. B. Allen and L. R. Rabiner, "A unified approach to short-time Fourier analysis and synthesis," *Proc. IEEE*, vol. 65, no. 11, pp. 1558–1564, Nov. 1977, doi: 10.1109/PROC.1977.10770.

[14] Z. Wang, L. Cao, Z. Zhang, X. Gong, Y. Sun, and H. Wang, "Short time Fourier transformation and deep neural networks for motor imagery brain computer interface recognition," *Concurr. Comput. Pract. Exp.*, vol. 30, no. 23, p. e4413, 2018, doi: 10.1002/cpe.4413.

[15] E. Yavuz and Ö. Aydemir, "Olfaction recognition by EEG analysis using wavelet transform features," in *2016 International Symposium on INnovations in Intelligent SysTems and Applications (INISTA)*, Aug. 2016, pp. 1–4. doi: 10.1109/INISTA.2016.7571827.

[16] B. L. Koley and D. Dey, "Detection of characteristic waves of sleep EEG by continuous wavelet transform," in *2012 NATIONAL CONFERENCE ON COMPUTING AND COMMUNICATION SYSTEMS*, Nov. 2012, pp. 1–5. doi: 10.1109/NCCCS.2012.6413025.

[17] J. Gu *et al.*, "Recent advances in convolutional neural networks," *Pattern Recognit.*, vol. 77, pp. 354–377, May 2018, doi: 10.1016/j.patcog.2017.10.013.

[18] S. Darvishi and A. Al-Ani, "Brain-Computer Interface Analysis using Continuous Wavelet Transform and Adaptive Neuro-Fuzzy Classifier," in *2007 29th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Aug. 2007, pp. 3220–3223. doi: 10.1109/IEMBS.2007.4353015.

[19] F. Yi, H. Li, and X. Jin, "Improved Classification Methods for Brain Computer Interface System," *Int. J. Comput. Netw. Inf. Secur.*, vol. 4, no. 2, pp. 15–21, Mar. 2012, doi: 10.5815/ijcnis.2012.02.03.

[20] S. Hu, Q. Tian, Y. Cao, J. Zhang, and W. Kong, "Motor imagery classification based on joint regression model and spectral power," *Neural Comput. Appl.*, vol. 23, no. 7, pp. 1931–1936, Dec. 2013, doi: 10.1007/s00521-012-1244-3.

[21] T.-U. Jang, B. M. Kim, Y.-M. Yang, W. Lim, and D.-H. Oh, "Motor-imagery EEG signal classification using position matching and vector quantisation," *Int. J. Telemed. Clin. Pract.*, vol. 1, no. 4, pp. 306–313, Jan. 2016, doi: 10.1504/IJTMCP.2016.078426.

[22] S. V. Eslahi, N. J. Dabanloo, and K. Maghooli, "A GA-based feature selection of the EEG signals by classification evaluation: Application in BCI systems." arXiv, Jan. 16, 2019. doi: 10.48550/arXiv.1903.02081.

[23] P. Kant, J. Hazarika, and S. H. Laskar, "Wavelet transform based approach for EEG feature selection of motor imagery data for braincomputer interfaces," in *2019 Third International Conference on Inventive Systems and Control (ICISC)*, Jan. 2019, pp. 101–105. doi: 10.1109/ICISC44355.2019.9036445.

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